CAPE DC/DC Modelling Delivery

Flyback Converter: State-Space Averaged Modelling

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Problem Setup

The flyback converter derives from the buck-boost topology with a transformer providing galvanic isolation. The magnetizing inductance L_M stores and transfers energy. The state vector is $\mathbf{x} = [i_g,\, v]^T$ (magnetizing current and output voltage), the control input is $u \in \{0, 1\}$, and the disturbance is $e = v_g$.

Ideal transformer relations: $v_P = v_S/n, \quad i_S = i_P/n.$

Question 1: Two Topologies (1p)

Topology 1: $u = 1$ (Q_1 ON, D_1 OFF) – interval dT_S

When Q_1 conducts, current flows through R_g and the primary winding, storing energy in L_M. The dot convention causes D_1 to be reverse-biased (secondary open-circuited). The capacitor C alone supplies the load R.

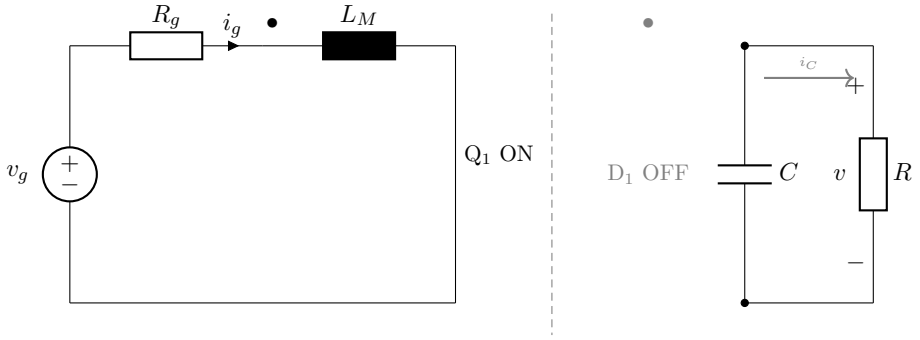


Figure 1: Topology for $u = 1$: Q_1 ON, D_1 OFF. Primary stores energy; secondary disconnected. Capacitor C discharges through R.

From KVL on the primary loop and KCL at the output node:

$$L_M \frac{di_g}{dt} = -R_g\, i_g + v_g \tag{1}$$

$$C\frac{dv}{dt} = -\frac{v}{R} \tag{2}$$

The two equations are decoupled: the primary side only sees R_g and L_M, while the secondary side has C discharging into R.

Topology 2: $u = 0$ (Q_1 OFF, D_1 ON) – interval $d'T_S$

When Q_1 turns off, the magnetizing current cannot change instantaneously. The voltage across L_M reverses, forward-biasing D_1. Energy transfers to the secondary. The reflected output voltage v/n appears across L_M (primary-referred), and the current i_g/n (secondary-referred) charges C and feeds R.

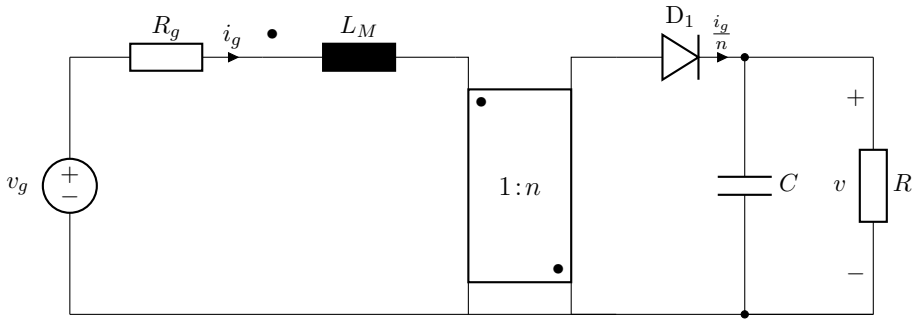


Figure 2: Topology for $u = 0$: Q_1 OFF, D_1 ON. Energy transfers from L_M to secondary via transformer. Current i_g/n charges C and feeds R.

From KVL on the primary loop (with reflected voltage) and KCL at the output:

$$L_M \frac{di_g}{dt} = -R_g\, i_g - \frac{v}{n} + v_g \tag{3}$$

$$C \frac{dv}{dt} = \frac{i_g}{n} - \frac{v}{R} \tag{4}$$

Now both state variables are coupled through the transformer turns ratio n.

Question 2: State Equations and Averaged Model (2p)

Matrix form for each topology

For $u = 1$ (Eqs. 1–2):

$$\dot{\mathbf{x}} = A_1\,\mathbf{x} + E_1\,e, \qquad y = C_1\,\mathbf{x} \tag{5}$$

$$A_1 = \begin{bmatrix} -\dfrac{R_g}{L_M} & 0 \\ 0 & -\dfrac{1}{RC} \end{bmatrix}, \quad E_1 = \begin{bmatrix} \dfrac{1}{L_M} \\ 0 \end{bmatrix}, \quad C_1 = \begin{bmatrix} 0 & 1 \end{bmatrix}, \quad F_1 = 0$$

For $u = 0$ (Eqs. 3–4):

$$\dot{\mathbf{x}} = A_2\,\mathbf{x} + E_2\,e, \qquad y = C_2\,\mathbf{x} \tag{6}$$

$$A_2 = \begin{bmatrix} -\dfrac{R_g}{L_M} & -\dfrac{1}{nL_M} \\ \dfrac{1}{nC} & -\dfrac{1}{RC} \end{bmatrix}, \quad E_2 = \begin{bmatrix} \dfrac{1}{L_M} \\ 0 \end{bmatrix}, \quad C_2 = \begin{bmatrix} 0 & 1 \end{bmatrix}, \quad F_2 = 0$$

Note: $E_1 = E_2$, $C_1 = C_2$, $F_1 = F_2 = 0$.

Averaged nonlinear model

Replacing the binary u by the continuous duty cycle d (with $d' = 1 - d$):

$$\dot{\mathbf{x}} = \big(A_1\,\mathbf{x} + E_1\,e\big)\cdot d + \big(A_2\,\mathbf{x} + E_2\,e\big)\cdot d'$$

Expanding and simplifying with $d + d' = 1$:

$$\boxed{\begin{aligned} L_M\frac{di_g}{dt} &= -R_g\,i_g - \frac{d'}{n}\,v + v_g \qquad (7)\\ C\frac{dv}{dt} &= \frac{d'}{n}\,i_g - \frac{v}{R} \qquad (8) \end{aligned}}$$

This is the **averaged nonlinear model**. The bilinear terms $d'\cdot v$ and $d'\cdot i_g$ make it nonlinear. Output: $y = v$.

Question 3: Steady-State Values (2.5p)

In steady state all derivatives vanish. Setting $d = D$, $d' = D' = 1 - D$:

$$0 = -R_g\, I_g - \frac{D'}{n}\, V + V_g \tag{9}$$

$$0 = \frac{D'}{n}\, I_g - \frac{V}{R} \tag{10}$$

From (10):

$$I_g = \frac{nV}{D'R} \tag{11}$$

Substituting into (9):

$$0 = -R_g \cdot \frac{nV}{D'R} - \frac{D'}{n}\, V + V_g \quad \Longrightarrow \quad V \cdot \frac{n^2 R_g + D'^2 R}{nD'R} = V_g$$

$$\boxed{V = \frac{nD'R}{n^2 R_g + D'^2 R}\, V_g} \tag{12}$$

$$\boxed{I_g = \frac{n^2}{n^2 R_g + D'^2 R}\, V_g} \tag{13}$$

Verification: lossless case ($R_g = 0$)

$$R_g = 0: \qquad V = \frac{n\, V_g}{D'} = \frac{n\, V_g}{1 - D}, \qquad I_g = \frac{n^2 V_g}{D'^2 R}$$

This recovers the standard flyback voltage gain $V/V_g = n/(1 - D)$.

Question 4: Linearized Model ($R_g = 0$) (2p)

From now on $R_g = 0$. The averaged model becomes:

$$L_M \frac{di_g}{dt} = -\frac{d'}{n} v + v_g \tag{14}$$

$$C \frac{dv}{dt} = \frac{d'}{n} i_g - \frac{v}{R} \tag{15}$$

Steady-state values with $R_g = 0$:

$$V = \frac{nV_g}{D'}, \qquad I_g = \frac{nV}{D'R} = \frac{n^2 V_g}{D'^2 R}$$

Perturbation and linearization

Split each variable into DC + small-signal perturbation:

$$i_g = I_g + \hat{i}_g, \quad v = V + \hat{v}, \quad d = D + \hat{d}, \quad d' = D' - \hat{d}, \quad v_g = V_g + \hat{v}_g$$

Inductor equation:

$$L_M \frac{d\hat{i}_g}{dt} = -\frac{(D' - \hat{d})}{n}(V + \hat{v}) + (V_g + \hat{v}_g)$$

Expanding, cancelling the steady-state part $\left(-\frac{D'}{n}V + V_g = 0\right)$, and neglecting the second-order term $\hat{d} \cdot \hat{v}$:

$$L_M \frac{d\hat{i}_g}{dt} = -\frac{D'}{n}\hat{v} + \frac{V}{n}\hat{d} + \hat{v}_g$$

Capacitor equation: analogous procedure, neglecting $\hat{d} \cdot \hat{i}_g$:

$$C \frac{d\hat{v}}{dt} = \frac{D'}{n}\hat{i}_g - \frac{\hat{v}}{R} - \frac{I_g}{n}\hat{d}$$

Linearized small-signal model

$$\boxed{L_M \frac{d\hat{i}_g}{dt} = -\frac{D'}{n}\,\hat{v} + \frac{V}{n}\,\hat{d} + \hat{v}_g} \tag{16}$$

$$\boxed{C \frac{d\hat{v}}{dt} = \frac{D'}{n}\,\hat{i}_g - \frac{\hat{v}}{R} - \frac{I_g}{n}\,\hat{d}} \tag{17}$$

Matrix form: $\dot{\hat{\mathbf{x}}} = A_{eq0}\,\hat{\mathbf{x}} + K_{eq0}\,\hat{d} + E_{eq0}\,\hat{e}, \quad \hat{y} = C_{eq0}\,\hat{\mathbf{x}}$

$$A_{eq0} = \begin{bmatrix} 0 & -\dfrac{D'}{nL_M} \\ \dfrac{D'}{nC} & -\dfrac{1}{RC} \end{bmatrix}, \quad K_{eq0} = \begin{bmatrix} \dfrac{V}{nL_M} \\ -\dfrac{I_g}{nC} \end{bmatrix}, \quad E_{eq0} = \begin{bmatrix} \dfrac{1}{L_M} \\ 0 \end{bmatrix}$$

$$C_{eq0} = \begin{bmatrix} 0 & 1 \end{bmatrix}, \qquad K'_{eq0} = 0, \qquad F_{eq0} = 0$$

Question 5: Transfer Functions (2.5p)

Applying the Laplace transform to (16)–(17):

$$sL_M\,\hat{I}_g(s) = -\frac{D'}{n}\hat{V}(s) + \frac{V}{n}\hat{D}(s) + \hat{V}_g(s) \tag{18}$$

$$sC\,\hat{V}(s) = \frac{D'}{n}\hat{I}_g(s) - \frac{\hat{V}(s)}{R} - \frac{I_g}{n}\hat{D}(s) \tag{19}$$

Common **characteristic polynomial** (denominator of all transfer functions):

$$\boxed{\Delta(s) = L_M C\,s^2 + \frac{L_M}{R}\,s + \frac{D'^2}{n^2}} \tag{20}$$

5.1 Input-to-output: $G_S(s) = \hat{V}(s)/\hat{V}_g(s)\big|_{\hat{D}=0}$

Set $\hat{D}(s) = 0$ in (18): $\hat{I}_g = \big[\hat{V}_g - (D'/n)\hat{V}\big]/(sL_M)$. Substitute into (19) and solve for $\hat{V}/\hat{V}_g$:

$$\boxed{G_S(s) = \frac{D'/n}{\Delta(s)}} \tag{21}$$

5.2 Control-to-output voltage: $G_C(s) = \hat{V}(s)/\hat{D}(s)\big|_{\hat{V}_g=0}$

Set $\hat{V}_g(s) = 0$. From (18): $\hat{I}_g = \big[-(D'/n)\hat{V} + (V/n)\hat{D}\big]/(sL_M)$.

Substitute into (19), multiply through by sL_M, and collect:

$$\Delta(s)\;\hat{V} = \left[\frac{D'V}{n^2} - \frac{sL_M I_g}{n}\right]\hat{D}$$

Using $I_g = nV/(D'R)$ gives $L_M I_g/n = L_M V/(D'R)$, and $D'V/n^2 = V_g/n$:

$$\boxed{G_C(s) = \frac{\dfrac{V_g}{n}\left(1 - s\,\dfrac{n^2 L_M}{D'^2 R}\right)}{\Delta(s)}} \tag{22}$$

This exhibits a **right-half-plane (RHP) zero** at $\omega_z = D'^2R/(n^2L_M)$, which limits the achievable closed-loop bandwidth.

5.3 Control-to-inductor current: $G_{Ci}(s) = \hat{I}_g(s)/\hat{D}(s)\big|_{\hat{V}_g=0}$

Set $\hat{V}_g(s) = 0$. From (19): $\hat{V} = \big[(D'/n)\hat{I}_g - (I_g/n)\hat{D}\big]/(sC + 1/R)$.

Substitute into (18) and solve. The numerator simplifies using $I_g = nV/(D'R)$:

$$\frac{V}{nR} + \frac{D'I_g}{n^2} = \frac{V}{nR} + \frac{V}{nR} = \frac{2V}{nR}$$

$$\boxed{G_{Ci}(s) = \frac{\dfrac{V_g}{D'R}\,(sRC + 2)}{\Delta(s)}} \tag{23}$$

This has a **left-half-plane (LHP) zero** at $\omega_z = 2/(RC)$, making the current loop minimum-phase and easier to stabilize (motivating current-mode control).

Summary and symbolic verification

All results verified with SymPy (Python symbolic math). Summary:

Transfer function	Numerator	Key feature
$G_S(s) = \hat{V}/\hat{V}_g$	D'/n	No finite zeros
$G_C(s) = \hat{V}/\hat{D}$	$\frac{V_g}{n}\left(1 - \frac{n^2 L_M}{D'^2 R}\, s\right)$	RHP zero
$G_{Ci}(s) = \hat{I}_g/\hat{D}$	$\frac{V_g}{D'R}(sRC + 2)$	LHP zero

Common denominator: $\Delta(s) = L_M C s^2 + (L_M/R)s + D'^2/n^2$

Natural frequency: $\omega_0 = \dfrac{D'}{n\sqrt{L_M C}}$, Quality factor: $Q = \dfrac{D'R}{n}\sqrt{\dfrac{C}{L_M}}$